Cross-Validated Tuning of Logistic Regression's Regularization Parameter C
An Empirical Study Across Dataset Structural Conditions

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1. Introduction

Cross-validation based hyperparameter tuning is standard in machine learning, yet whether tuning logistic regression's regularization parameter C actually improves over *scikit-learn*'s default of C=1.0 remains underexplored given that the selection process itself introduces variance through random data splits. This project aims to answer the following question: does cross-validated tuning of logistic regression's regularization parameter C reliably improve classification accuracy, and under what dataset conditions, if any, does it outperform the default C=1.0?

To answer this question, we conduct a two-phase empirical study. In Phase I, we generate synthetic datasets under controlled conditions while varying three structural properties: the feature-to-sample ratio, the class imbalance and the feature redundancy, across ten intensity levels each. In Phase II, we analyze ten real-world datasets, four of which serve as baselines with moderate structural properties and six of which are selected specifically for their extreme values of the structural properties examined in Phase I. In both phases, all experiments are averaged across 20 random seeds to obtain stable estimates of tuning gain. Logistic regression was selected because it has a single main regularization parameter C, allowing us to isolate the effect of tuning one hyperparameter in a controlled and interpretable setting.

Note: throughout this report, we use *scikit-learn*'s C notation, where $C = 1/\lambda$ and λ denotes the L2 regularization strength as defined in the course.

2. Related Work

Both the design and the interpretation of this study are motivated by prior work.

Krstajic et al. (2014) show that a single cross-validation run is not reliable for model selection, as the optimal parameter chosen can vary depending on how the data is split. They propose repeating cross-validation multiple times and averaging the results to obtain more robust estimates. This directly motivates our use of 20 random seeds per experiment: instead of reporting tuning gain from a single train-test split, we average across 20 different splits to get stable estimates. Their work also helps explain the high variability when it comes to the gains for every seed, suggesting that instability in CV-based parameter selection is a fundamental property of the method itself, not a problem specific to our experimental setup.

Uddin and Lu (2024) study the relationship between dataset structural properties and machine learning performance across 200 tabular datasets and five classifiers. They find that dataset characteristics interact with the accuracy of classifiers that aren't tree-based, including logistic regression. Our study extends this by asking whether those same structural properties are associated with the gain obtained through CV-based hyperparameter tuning. However, Uddin and Lu needed 200 datasets to detect significant relationships, whereas our statistical power and correlation analysis is limited due to only using ten datasets.

Oreski et al. (2017) show that across 128 datasets, structural dataset characteristics, including feature noise, data sparsity and inter-feature correlation, determine which feature selection technique performs best. We extend this idea to hyperparameter tuning, asking whether structural properties are associated with the benefit of tuning C in logistic regression.

3. Methods

3.1 Shared Experimental Pipeline

Both phases follow an identical pipeline: each dataset (X, y) is split into 80% training and 20% test sets. Features are standardized using a *StandardScaler* fitted on the training set only and then applied to both sets, so that test set information does not influence the scaling parameters.

Two logistic regression models are then trained on the standardized training data. The first is the default model, trained with *scikit-learn*'s default C=1.0. The second is the CV-tuned model, for which the optimal value of C is selected via five-fold cross-validation on the training set alone. Six candidate values are considered: $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$, from strong to weak regularization. The C with the highest mean validation accuracy across the five folds is selected, and a final model is retrained on the full training set with that C before evaluating it on the test set.

For a given experiment, we define: $\text{gain} = \text{accuracy}_{\text{tuned}} - \text{accuracy}_{\text{default}}$

A positive gain means tuning improved over the default. A negative gain means cross-validation selection noise led to a suboptimal C being chosen, causing the tuned model to perform worse than the default on the test set. To obtain stable estimates of tuning gain, the entire pipeline is repeated 20 times using seeds 0 through 19, each controlling a different random train-test split. The mean gain across the 20 seeds and its standard deviation are reported. The same 20 seeds are used across all datasets and conditions.

3.2 Phase I: Synthetic Controlled Experiments

Phase I studies the controlled relationship between each structural dataset property and tuning gain. Synthetic datasets are generated using *scikit-learn's make_classification*, so that a single property is varied across ten intensity levels while the others stay constant, a level of control not possible with real-world data. Three structural properties are examined. For each property and intensity level pair, the pipeline described in Section 3.1 is applied across all 20 seeds, resulting in 200 individual experiments per property.

3.2.1 Feature-to-Sample Ratio (p/n)

The feature-to-sample ratio p/n is varied by fixing the number of features at p=20 and varying the number of samples from 1000 down to 80, resulting in ratios from p/n=0.02 to p/n=0.25. All other parameters are held constant and moderate: 8 informative features, 1 redundant feature, 0 repeated features, balanced classes and a class separation of 1.0, and each dataset is generated with *random_state=42* so the dataset itself is identical across all 20 seeds.

This is motivated by the idea that logistic regression may be more sensitive to regularization when features outnumber samples, since it is associated with higher overfitting risk where tuning C is expected to matter more.

3.2.2 Class Imbalance

The class imbalance ratio is the minority class count divided by the majority class count, with lower values indicating greater imbalance. Class imbalance is varied by fixing the number of samples at 500 and features at 10, and varying the minority class proportion from 45% down to 1% across ten levels using *make_classification*'s weights parameter. All other parameters are kept constant: 8 informative features, 1 redundant feature and a class separation of 1.0.

This property is examined because severe class imbalance may affect tuning. For imbalanced datasets with very few minority class samples per fold, cross-validation estimates could become unstable, potentially leading to unreliable C selection.

3.2.3 Feature Redundancy as a Correlation Proxy

Feature redundancy is used as a proxy for inter-feature correlation. In *make_classification*, redundant features are linear combinations of the informative ones, directly increasing mean pairwise correlation. The number of redundant features is varied from 0 to 7, with the total fixed at 10 where possible, reduced to 9 and 8 in the two highest redundancy conditions since *scikit-learn* requires at least 2 informative features.

This property is examined because multicollinearity affects the logistic regression's weight estimates that regularization helps stabilize. We want to explore if tuning gain is larger in high-correlation settings.

3.3 Phase II: Real-World Datasets

Phase II applies the same pipeline to ten real-world datasets to determine whether the Phase I findings hold on real-world data. However, isolating a single structural property is not as straightforward in real datasets. Six datasets were therefore selected specifically for their extreme values of one structural property, while four baseline datasets with moderate properties serve as a reference group.

3.3.1 Baseline Datasets

The four baseline datasets, from *scikit-learn*'s built-in dataset collection, are: breast cancer (569 samples, 30 features), wine (178 samples, 13 features), iris (150 samples, 4 features) and digits (1797 samples, 64 features). These datasets were selected because they require minimal preprocessing and none of them presents an extreme feature-to-sample ratio, class imbalance or unusually high or low inter-feature correlation. They are used as the reference to which the targeted datasets are compared against.

3.3.2 Targeted Datasets

Six OpenML datasets are each selected for an extreme value of a structural property:

- High p/n ratio: Colon cancer (62 samples, 2000 features, p/n = 32.26)
- Low p/n ratio: Pima Indians diabetes (768 samples, 8 features, p/n = 0.01)
- High class imbalance: Credit card fraud (minority class = 0.17%. Sampled to 50,000 records using *numpy.random.default_rng(42)* to maintain class distribution)
- Low class imbalance: Banknote authentication (minority class = 44.8%)
- High feature correlation: Abalone (physical measurements of sea snails such as length, diameter, height, weight, all related to animal size, $\bar{r} = 0.68$. Target binarised at median ring count: above = old, at or below = young)
- Low feature correlation: Madelon (artificially engineered dataset, $\bar{r} = 0.02$)

Although madelon is artificially engineered and not naturally occurring data, it was included because it was the lowest correlation dataset we could find, which was needed for our targeted study design. The mean feature correlation is the mean of the absolute values of the upper triangle of the Pearson correlation matrix, computed after removing constant features.

3.3.3 Preprocessing

All OpenML datasets are loaded using *scikit-learn's fetch_openml.* Categorical features are encoded as integers using a *LabelEncoder*. Columns consisting entirely of missing values are dropped, and any row containing at least one missing value is removed along with its label. Class labels are re-encoded as consecutive integers starting from zero. The datasets are then standardized as described in Section 3.1.

3.3.4 Phase II Pipeline Differences from Phase I

Two modifications are applied in Phase II. First, the train-test split uses stratified sampling to keep class distribution across all 20 seeds, which is important for credit fraud, where the minority class represents only 0.17% of the data. Without stratification, some random splits could contain no fraud cases in the test set. If stratification fails, the split falls back to unstratified sampling. Second, any C value that causes a cross-validation exception is skipped, with the best C being 1.0 by default if all six candidates fail.

3.4 Statistical Analysis

3.4.1 One-Sample t-Test for Gain Significance

For each dataset, a one-sample t-test determines if the mean tuning gain across 20 seeds is statistically distinguishable from zero (H_0: $\mu_{gain} = 0$, two-tailed, $\alpha=0.05$). The standard error and 95% confidence interval are computed as:

$SE = \sigma_{gain} / \sqrt{20}$

$CI = mean_{gain} \pm t_crit \times SE$, where $t_crit \approx 2.093$ ($df = 19$)

If the confidence interval contains zero, the gain is not statistically significant. For credit fraud, all 20 seeds produced a gain of exactly zero. Therefore, no t-test could be computed and the result is reported as 0.00%.

3.4.2 Correlation Analysis

To determine if any structural property is associated with tuning gain, we computed the Pearson and the Spearman correlation coefficients between each structural property and the mean gain across the ten datasets. Spearman is chosen because it is rank-based, making it more robust to outliers in the case of non-linear relationships, and colon cancer is a potential outlier with p/n=32.26, so it could affect the Pearson estimate. For each Pearson r, a 95% confidence interval is computed using the Fisher z-transformation.

Effect sizes are classified independently of p-values as: strong ($|r| \geq 0.7$), moderate ($|r| \geq 0.5$), weak ($|r| \geq 0.3$) or very weak ($|r| < 0.3$). With only n=10 datasets, the analysis is underpowered: almost no correlation will reach statistical significance regardless of how strong it is. Therefore, we report both the effect size and the p-values for a clearer picture.

3.4.3 Sign Test

The sign test asks a simple question: across the ten datasets, does tuning help more often than it hurts? Under H_0, positive and negative gains are equally likely (p=0.5). The number of datasets with a positive mean gain is tested against this null hypothesis using a two-tailed binomial test at $\alpha=0.05$. This is an appropriate analysis, because it only considers the direction of the gain, not its magnitude, making it more robust than other metrics to the high variability observed across datasets.

4. Results

4.1 Phase I: Synthetic Controlled Experiments

Figure 1. Mean Accuracy Gain by Dataset Structural Property - Synthetic Study

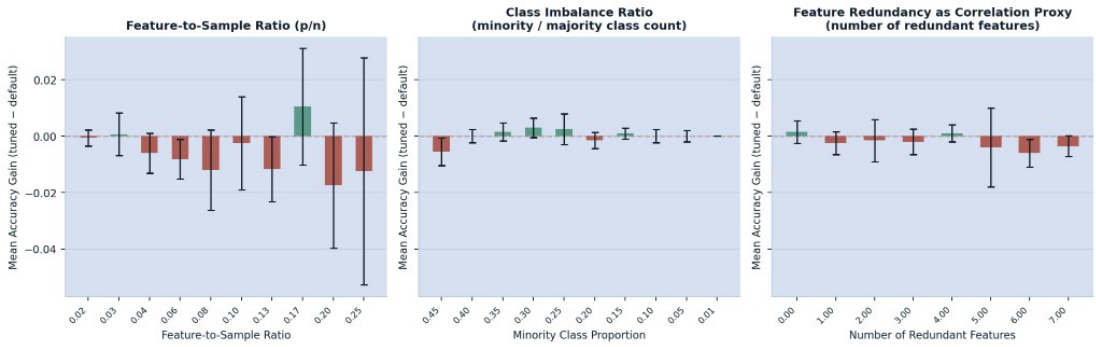


Figure 1 presents the mean tuning gain and its 95% confidence interval across all ten intensity levels for each structural property. The pattern is consistent across all three panels: mean gains remain close to zero at every intensity level, and the standard deviation consistently exceeds the mean gain: variability dominates any tuning effect.

For the feature-to-sample ratio, gains range from -1.75% to +1.04% with no monotonic trend. Variability grows with higher ratios, reaching ±9.19% at p/n=0.25. This is consistent with Krstajic et al. (2014): confidence intervals get wider due to model instability when the number of samples is small relative to the number of features.

For class imbalance, gains are negligible, ranging from -0.55% to +0.30%, showing no trend. At the most extreme 99-1 class imbalance case, the gain is exactly zero.

For feature redundancy, gains fluctuate around zero across all eight levels, ranging from +0.15% to -0.60%. A slight negative trend appears at higher redundancy levels but stays within the noise. Higher feature correlation does not make tuning C more beneficial, contrary to what we expected.

4.2 Phase II: Real-World Datasets

Table 1 summarises the six targeted datasets and their observed mean gains.

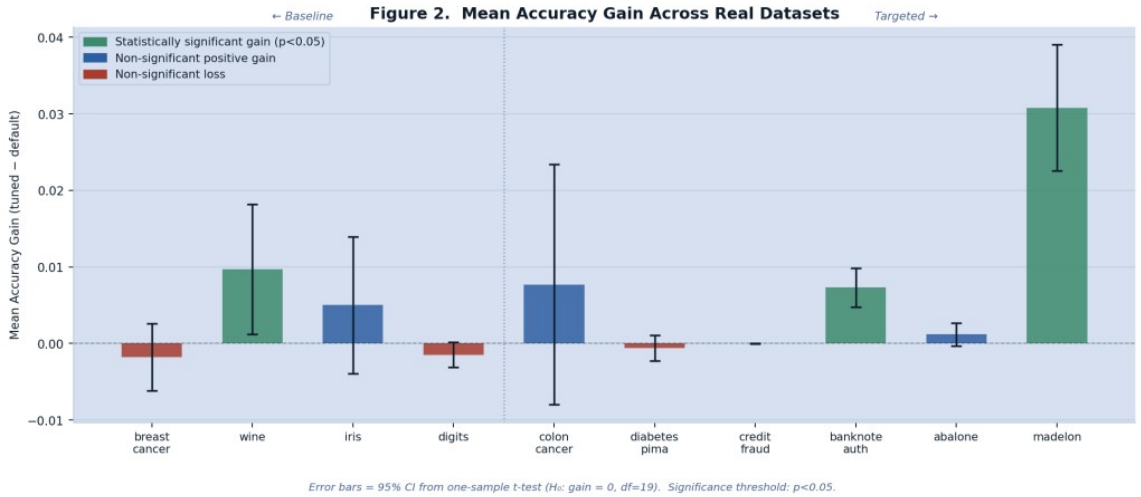
Table 1. Targeted Datasets: Structural Properties and Mean Accuracy Gain

Dataset	Target Property	Property Value	Mean Gain
Colon cancer	High feature/sample ratio	p/n = 32.26	+0.77%
Diabetes pima	Low feature/sample ratio	p/n = 0.01	−0.06%
Credit fraud	High class imbalance	minority = 0.18%	+0.00%
Banknote auth	Low class imbalance	minority = 44.8%	+0.73%
Abalone	High feature correlation	$\bar{r}$ = 0.68	+0.12%
Madelon	Low feature correlation	$\bar{r}$ = 0.02	+3.08%

Each dataset selected for an extreme value of one structural property. Significance from one-sample t-test (H_0: gain = 0, df=19, α=0.05).

Only two targeted datasets show statistically significant gains at α=0.05: banknote auth (+0.73%, p<0.001) and madelon (+3.08%, p<0.001).

Figure 2 presents the mean tuning gain and its 95% confidence interval for each of the ten real datasets.



Madelon shows the largest gain of +3.08% ($p<0.001$). It is an artificially designed classification problem with very low feature correlation ($\bar{r}=0.02$), which may explain why regularization tuning matters more in this case. Banknote auth shows a +0.73% gain with a very narrow confidence interval, indicating consistent benefit across all 20 seeds. Wine shows a +0.97% ($p=0.028$).

At the other extreme, credit fraud shows zero gain: both models achieve 99.9% accuracy regardless of C, since both simply predict the majority class when class imbalance gets this severe. This makes credit fraud uninformative for our purposes: when both models default to predicting the majority class, accuracy can no longer distinguish between values of C. Breast cancer (-0.18%) and digits (-0.15%) show minor accuracy losses, suggesting CV selection noise led to a suboptimal C. Colon cancer, despite its extreme p/n=32.26, shows a non-significant +0.77% with a wide confidence interval, likely due to high variability instead of a genuine benefit.

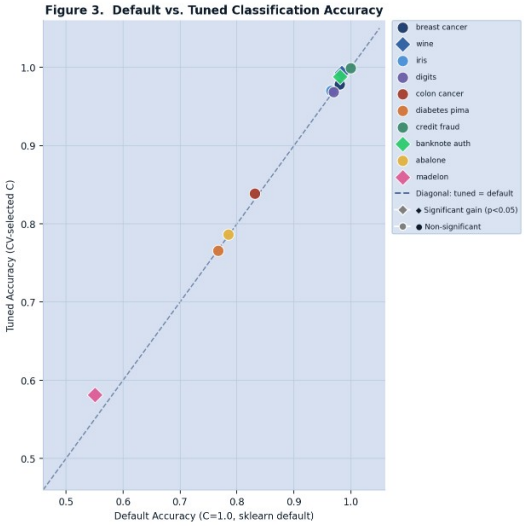


Figure 3 shows the same results as a scatter plot of default vs. tuned accuracy, where points above the diagonal indicate a positive gain and points below indicate accuracy degradation. All datasets cluster near the diagonal, confirming that the accuracy achieved by tuned and default models is nearly the same in most cases.

4.3. Statistical Analysis

4.3.1 Sign Test

Six out of ten datasets show a positive mean gain and four show a negative or zero gain. A two-tailed binomial test (H_0: p=0.5) results in p=0.754. There is therefore no evidence that CV tuning systematically tends to help more than it hurts, since getting 6/10 positive could very well be by chance.

4.3.2 Correlation Analysis

Table 2 presents Pearson and Spearman correlations between each structural property and mean tuning gain. All five properties show weak to very weak correlations, and none reaches statistical significance.

Table 2. Pearson & Spearman Correlations: Structural Dataset Properties vs. Mean Accuracy Gain

Property	Pearson r	95% CI	p-value	Spearman ρ	p-value (ρ)	Strength
Feature/Sample Ratio	+0.074	[-0.583, +0.672]	0.839	+0.430	0.214	Very weak
Class Imbalance Ratio	+0.355	[-0.354, +0.805]	0.314	+0.347	0.327	Weak
Mean Feature Correlation	-0.301	[-0.782, +0.406]	0.399	-0.067	0.855	Weak
N Samples	-0.194	[-0.734, +0.496]	0.592	-0.188	0.603	Very weak
N Features	+0.281	[-0.424, +0.774]	0.432	+0.085	0.815	Very weak

95% CI via Fisher z-transformation. All correlations non-significant, therefore the results are exploratory given n=10.

The strongest Pearson correlations are with class imbalance ratio (r=+0.355, p=0.314) and mean feature correlation (r=-0.301, p=0.399). The feature-to-sample ratio has a Pearson correlation of almost zero (r=+0.074, p=0.839). Spearman correlations are broadly consistent, with one exception: feature/sample ratio shows a much higher Spearman ρ=+0.430 than its Pearson r=+0.074. This is likely because colon cancer's extreme p/n=32.26 distorts the Pearson estimate, while Spearman's rank-based approach handles that outlier better.

With only n=10 datasets, the analysis is very underpowered: even a true correlation could go by undetected. These results are therefore exploratory: the absence of significant correlations is consistent with both a genuine null effect and insufficient power to detect weak relationships.

5. Discussion

The results consistently point toward one conclusion: CV tuning of logistic regression's C does not reliably improve classification accuracy across the dataset conditions examined. This holds across both phases.

In Phase I, no condition produced a mean gain large relative to its variability. The standard deviation of the gain consistently exceeded the mean. This is directly consistent with Krstajic et al. (2014), who show that CV-based model selection is noisy: even averaging over 20 seeds could not counter that effect.

Only three real-world datasets show significant gains: wine, banknote auth and madelon. However, we interpret these as isolated cases instead of a pattern, since the correlations between the dataset structural properties we studied and the mean gains are all very weak to weak.

It is important to note that we cannot conclusively establish that no relationship exists: with only ten datasets, the analysis is too underpowered. Uddin and Lu (2024) needed 200 datasets to detect significant property-performance relationships. Replicating the experiment at a larger scale would be needed to draw definitive conclusions.

Two other limitations are worth noting regarding our experimental design. First, C is selected using a single five-fold CV run per seed rather than repeated CV as recommended by Krstajic et al. (2014), which contributes to high variability per seed. Second, the candidate set of C values is limited to six: a finer search could improve results.

6. Conclusion

This study evaluated whether CV tuning of logistic regression's C reliably improves classification accuracy across a range of dataset structural conditions. Using a two-phase design, 30 synthetic conditions and 10 real datasets, each averaged over 20 seeds, we find no evidence of a reliable tuning benefit in the vast majority of conditions tested.

In Phase I, gains remained near zero at every intensity level: variability across seeds consistently exceeded the mean gain. In Phase II, only 3 out of 10 datasets showed significant gains, and the sign test confirmed that getting 6 out of 10 positive gains is not significant ($p=0.754$). No structural property predicted when tuning would help.

Based on our results, *scikit-learn's* default C=1.0 appears to be a strong and stable baseline in most conditions. CV tuning introduces selection noise that can occasionally degrade performance, and we found no consistent accuracy improvement that would justify the additional computational cost.

Future work should replicate this at larger scale for sufficient statistical power, extend to other classifiers and hyperparameters and consider repeated CV within each seed to reduce selection noise.

7. Reproducibility

All code is publicly available at https://github.com/melanie-ghaby/cv-tuning-logreg-C-study, containing four Python scripts: *phase1_synthetic_data.py, phase2_real_datasets.py, descriptive_statistics.py* and *visualizations.py*, along with all intermediate CSV outputs.

Phase I synthetic datasets are reproducible via *random_state=42* in *make_classification*. Phase II real datasets are publicly available from *scikit-learn* and OpenML. The credit fraud subsample uses *numpy.random.default_rng(42)*. All experiments use seeds 0 through 19, fixed and identical across all conditions.

Experiments were run on Windows with an Intel Core i7. Phase I completes in under one minute, Phase II in approximately five minutes. Software versions: Python 3.14.0, scikit-learn 1.8.0, numpy 2.4.4, pandas 3.0.2, scipy 1.17.1.

8. References

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